Table I: Some Popular Open Source Development Boards

Specifications		Banana Pi	BeagleBone Black	Cosmic + Board	Cubieboard 3	CuBox-i4Pro	Game-Stick	Humming-Board i2eX	Intel Galileo	Intel NUC Board DE-3815TYBE	MarsBoard RK3066	MiraBox	Novena	ODROID-U3	Radxa Rock Lite	Raspberry Pi B+	RIoTboard	RouterBOARD RB953GS-5HnT	UDOO Quad	VIA Springboard VAB-600
SoC		Allwinner A20	TI Sitara AM335x	Freescale Vybrid VF6xx	Allwinner A20	Freescale i.MX6 Quad	Amlogic 8726-MX	Freescale i.MX6 Dual	Intel Quark SoC X1000	Intel Atom E3815	Rockchip RK3066	Marvell Armada 370	Freescale i.MX6 Quad	Samsung Exynos 4 Quad	Rockchip RK3188	Broadcom BCM2835	Freescale i.MX6 Solo	Qualcomm Atheros QCA9558	Freescale i.MX6 Quad/ Atmel SAM3X8E	WonderMedia WM8950
CPU	Architecture	ARM Cortex-A7	ARM Cortex-A8	ARM Cortex-A5/ARM Cortex-M4	ARM Cortex-A7	ARM Cortex-A9	ARM Cortex-A9	ARM Cortex-A9	x86 Quark	x86-64 Bay Trail-I	ARM Cortex-A9	ARMv7	ARM Cortex-A9	ARM Cortex-A9	ARM Cortex-A9	ARM11	ARM Cortex-A9	MIPS 74Kc	ARM Cortex-A9/ARM Cortex-M3	ARM Cortex-A9
	Cores			2 (1 + 1)															5 (4 + 1)	
	Frequency	1 GHz	1 GHz	500 MHz/167 MHz	1 GHz	1 GHz	1 GHz	1 GHz	400 MHz	1.46 GHz	1.6 GHz	1.2 GHz	1.2 GHz	1.7 GHz	1.6 GHz	700 MHz	1 GHz	720 MHz	1 GHz/?	800 MHz
GPU		Mali-400MP2	PowerVR SGX530	?	Mali-400MP2	Vivante GC2000 + GC335 + GC320	Mali-400	Vivante GC2000 + GC355 + GC320	N/A	Intel HD Graphics	Mali-400MP4	N/A	Vivante GC2000	Mali-400MP4 @440 MHz	Mali-400MP4	Broadcom VideoCore IV	Vivante GC880 + GC320	N/A	Vivante GC2000 + GC355 + GC320	Mali-400
RAM	Size	1 GB	512 MB	256 MB	2 GB	2 GB	1 GB	1 GB	256 MB	1 SO-DIMM socket	1-2 GB	1 GB	4 GB SO-DIMM	2 GB	1 GB	512 MB	1 GB	128 MB	1 GB	1 GB
	Data Rate (MT/s)	?	?	?			?			up to 1066	?				?	?	?	?	?	
	Data Path Width (bits)	?		?	?		?		?		?			?	?	?	?	?	?	
	Type	DDR3	DDR3L	DDR3	DDR3	DDR3	DDR3	DDR3	DDR3	DDR3L	DDR3	DDR3L	DDR3	LPDDR2	DDR3	?	DDR3	DDR2	DDR3	DDR3
PCIe		No	No	No	No	No	No	1 mini	1 mini	1 mini	No	1 mini	1 mini	No	No	No	No	2 mini	No	1 mini
USB	2.0									2 + 3 header										
	3.0	No	No	No	No	No	No	No	No		No		No	No	No	No	No	No	No	No
	Device	OTG	Device	OTG	OTG	No	No	?	Client	No	OTG	No	OTG	OTG	OTG	?	OTG	No	OTG	No
Storage	On Board	No	4 GB Flash	512 MB Flash	8 GB Flash	No	8 GB Flash	No	8 MB Flash + 8 KB EEPROM	4 GB eMMC	4 GB Flash	1 GB Flash	No	eMMC module opt.	4 GB Flash	No	4 GB Flash	128 MB Flash	No	4 GB eMMC
	Flash Slots	SD	microSD	microSD (SDHC)	microSD	microSD	microSD	microSD (UHS)	SD	No	microSD	microSD	SD	microSD	microSD (SDXC)	microSD	microSD and SD	No	microSD	microSD
	SATA	Yes	No	No	SATA 2.0	eSATA 2.0	No	mSATA	No	SATA 2.0	No	No	SATA 2.0	?	?	No	No	No	SATA	No
Networking	Ethernet	Gigabyte	10/100	10/100	Gigabyte	Gigabyte	Dock	Gigabyte		Gigabyte Ethernet (RTL8111)		2x GbE	1 GbE + 1 100Mbit				GbE	3x GbE (AR8327)	GbE	10/100 (VT6113)
	Wi-Fi	No	No	No	a/b/g/n(BCM4329)	b/g/n (BCM4329)	b/g/n	No	No	No	b/g/n RTL8188)	b/g/n (88W8787)	b/g/n	No	b/g/n (RTL8188)	No	No	a/n (QCA9558)	n (RT5370)	b/g/n opt. (AR9271)
Communication	Bluetooth	No	No	No	2.1 + EDR	2.1 + EDR	4.0 LE	No	No	No	No		Opt.	No	No	No	No	No	No	No
	I2C	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes	No	Yes	Yes
	SPI	Yes	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	Yes	Yes	Yes	No	Yes	Yes
Generic I/O	GPIO				Yes	?	?			Yes	?		Header	Yes				No		
	Analog	No	12-bit ADC	12-bit ADC, 12-bit DAC, PWM	?	No	No	No	12-bit ADC, 6 PWM	PWM	No	No	8x 12-bit ADC, 8x PWM, via FPGA	No	ADC, PWM	No	PWM	No	10-bit ADC, PWM	No
Other Interfaces		CSI, UART	CAN, UART	CAN, JTAG, OpenSDA, RS-232	IrDA, UART	CIR rx/tx		CIR rx, CSI-2, FlexCAN, UART	Arduino 1.0 headers, JTAG, 6x UART	SMBus	CIF, UART	JTAG	High-speed I/O expansion header (provides GPIO breakout)	UART	UART	UART	CSI, UART	RS-232, 2x SFP	Arduino 1.0 headers	CIR
Audiovisual Interfaces	Mic In	Pads	No	No	Header	No	No	Yes	N/A	3.5 mm	Yes	N/A	Yes	3.5 mm	?	No	Yes	N/A	Yes	3.5 mm
	Audio Out	Yes	HDMI	No	3.5 mm, HDMI, S/PDIF	Optical SPDI/F	HDMI	3.5mm jack, S/PDIF	N/A	3.5 mm	Yes	N/A	Yes	3.5 mm, HDMI	3.5 mm, HDMI, S/PDIF	Yes	3.5 mm, HDMI	N/A	Yes	3.5 mm
	HDMI	Yes	Yes	Micro HDMI			Yes	Yes	N/A	HDMI 1.4a	Yes	N/A	Yes	Yes		YesDVI compatible	Yes	N/A	Yes	Mini HDMI 1.4
	LVDS	Yes	No	No	No	No	No	LCD header	N/A	LCD header	RGB	N/A	Part of "eDP Adapter Board"	No	Yes	?	Yes	N/A	LCD header	DVO connector
	Other Video Out	Composite	No	No	VGA	No	No	No	N/A	VGA header	No	N/A	No	No	AV output	Composite video	?	N/A	No	No
Operating System		Linux Android	Linux Android	Linux MQX RTOS	Android	Linux Android	Android	Linux Android	Linux Windows	Linux Android BSD Windows	Linux Android	Linux	Linux	Linux Android	Ubuntu/ Linaro 13.09 Android	Linux FreeBSD NetBSD RISC OS Plan 9	Linux	RouterOS 5	Linux Android	Linux Android
Size (mm)		92 × 60	86.4 × 53.3	100 × 72	110 × 80	55 × 55 × 42	?	85.6 × 54	123.8 × 72	101.6 × 101.6	105x78	133 × 93 × 20	?	83 × 48	100 x 80 x 30	85.6 × 54.0 × 19.5	120 x 75	183 × 105 × 24	110 x 85	100 × 72 (Pico-ITX)
Weight		48 g	40 g	64.2 g	220 g (kit)	?	?	?	?	?	90 g	?	?	48 g	?	45 g	?	?	?	?
Input Voltage		5 V	5 V	5 V	5 V	5 V	5 V	5 V	7-15 V	12 V	5 V	5 V	16 V DC	5 V	5 V	5 V	5 V	8-30 V	6-15 V	12-24 V
Power Consumption		10 W supply, 1.15 W idle in Linux	1.05–2.3 W	5 W supply, 1.83 W typ.	?	15 W supply	?	10 W supply	?	36 W supply	10 W supply	15 W	1.6 W	?	?	3 W	5 W supply	25 W max	?	?
Power Source		micro USB	miniUSB or DC jack or via expansion header	Mini USB or DC jack	miniUSB or DC jack or LiPo batteries	DC jack	Micro USB	Micro USB	DC jack or PoE	DC jack	DC jack	DC jack	DC jack, or any 3 cell RC LiPo battery via optional charger module	?	DC jack or USB OTG input	Micro USB or GPIO header	DC jack	DC jack	DC jack	DC jack